

Overview

The system is designed to collect, unify, and analyze daily sales reports from multiple restaurant locations across Spain. Each branch exports daily CSV files containing sales data, operational parameters (opening hours, number of tables), and product-level transactions.

The main goal is to generate actionable business insights such as best-selling products, peak sales hours, and performance differences between locations.

Architecture

CSV Sources → ETL Pipeline → PostgreSQL Database → Analytics Layer → Insights / Reports

1. Data Source Layer

Input Sources

Each restaurant generates daily files containing:

- Sales receipts (bills)
- Line items per receipt (bill items)
- Timestamp of transactions
- Payment methods
- Total amounts
- Optional operational metadata (location-specific rules, hours, etc.)

Characteristics

- Heterogeneous file formats across locations
 - CSV / Excel-based exports from POS systems
 - End-of-day batch delivery
-

2. ETL Layer (Ingestion & Transformation)

2.1 File Collector

- Retrieves daily files from all restaurants
 - Sources can include:
 - shared folders
 - FTP/S3 storage
 - email uploads
 - Runs in batch mode (end-of-day processing)
-

2.2 Parser & Normalizer

Transforms raw Excel/CSV into structured entities:

- restaurants
- bills
- bill_items
- menu_items (optionally enriched)

Responsibilities:

- column mapping across different formats
 - data type normalization (timestamps, numeric fields)
 - cleaning missing/invalid values
 - standardizing product names and codes
-

2.3 Data Validation Layer

Ensures data integrity before storage:

- schema validation
 - duplicate prevention:
 - (`restaurant_id`, `bill_no`, `sold_at`) uniqueness constraint
 - consistency checks:
 - total bill = sum of bill items
 - detection of invalid values (negative quantities, null prices)
-

2.4 Data Loader

- Inserts cleaned data into PostgreSQL
 - Batch inserts for efficiency
 - UPSERT logic for safe reprocessing
-

3. Data Storage Layer (PostgreSQL)

Core Tables

- **restaurants** → restaurant metadata
 - **menu_items** → product catalog
 - **bills** → transaction headers (fact table)
 - **bill_items** → transaction details (fact table)
-

Data Model Type

This design follows a **dimensional (star-like) model**:

- Fact tables:
 - bills
 - bill_items
 - Dimension tables:
 - restaurants
 - menu_items
-

Indexing Strategy

- `bills(restaurant_id, sold_at)` → time-based analytics
- `bill_items(menu_item_id)` → product-level queries
- `bill_items(bill_id)` → fast joins between header and items

4. Analytics Layer (Business Intelligence Engine)

Purpose

Transform raw transactional data into actionable business insights.

Core Analytical Modules



Sales Performance

- revenue per restaurant
 - daily and hourly revenue trends
 - average ticket size per bill
-



Time-Based Analytics

- peak hours analysis
 - daily patterns (weekday vs weekend)
 - lunch vs dinner segmentation
-



Product Performance

- best-selling menu items
 - revenue contribution per product
 - product performance across locations
-



Location Comparison

- ranking restaurants by:
 - total revenue
 - number of transactions
 - average order value

(Optional) Anomaly Detection

- sudden drop in sales
 - unusual spikes in product demand
 - operational inconsistencies
-

5. Reporting Layer

Output Formats

Option A: SQL Views

- precomputed analytical views:
 - daily_sales_summary
 - product_performance_summary
 - hourly_sales_distribution
-

Option B: BI Dashboards

- tools such as:
 - Metabase
 - Apache Superset
 - Power BI
-

Option C: API Layer (optional extension)

REST endpoints for external access:

- `/top-products`
 - `/sales-by-hour`
 - `/restaurant-performance`
-

6. End-of-Day Processing Flow

[Restaurant CSV/Excel Files]



[File Collector Service]



[Parser + Normalization Layer]



[Validation & Cleaning]



[PostgreSQL Database]



[Analytics Queries]



[Reports / Dashboards]